

# Gaurav Kumar

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## Technical Skills

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**Languages:** Python, C++, Java, Rust, Golang, JavaScript, TypeScript  
**Frameworks:** Django, Django Rest Framework/, React, React Native, REST API, Node.js  
**Databases:** MySQL, PostgreSQL, MongoDB, Redis, GraphQL, Prometheus, Dynamo DB  
**Generative AI:** GANs, VAE, Diffusion Models, LLMs, Llama, Anthropic, OpenAI, RAG  
**Cloud Platforms:** GCP, AWS (EC2, S3, SQS, Lambda, Cloudwatch, ECR), Azure  
**Tools:** PowerBI, Tableau, Hadoop, Snowflake, LlamaIndex, Langchain

## Research Experience

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### Sutton Lab

Graduate Research Assistant

Columbia, SC

01/2024 – Present

- Working on leveraging generative diffusion models to propose molecular structures, addressing combinatorial scaling challenges and enhancing material structure prediction efficiency.
- Developed tree-based models to accurately predict JV Reverse Scan PCE of Perovskite Materials and worked on fine-tuning encoder based model on molecular structures for precise band gap predictions.

## Work Experience

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### AWS

Software Engineer

New York, NY

07/2024 – Present

- Designing, developing, and maintaining services that work with AWS Private Pricing Agreements (PPAs) to identify custom and pre-approved contract language, catch drafting errors, and auto-ingest contract data into AWS billing.
- Building and maintaining a term library and term engine that programmatically represents contract language and maps it to billing schemas.
- Building and maintaining an auto-SIM feature that automates tick creation for monetization teams.

### Fractal Analytics

Data Scientist

Mumbai, India

10/2020 – 05/2022

- Built a Customer Lifetime Value (CLV) model for a retail client, optimizing marketing spend across multiple channels and enhancing ROI by 1%, resulting in a \$10M increase in annual revenue.
- Developed and deployed an implicit collaborative filtering recommendation model and engineered a seasonal prioritization multiplier, resulting in a 7% (\$100k/month) increase in sales validated through A/B testing.
- Increased lead conversion rate by 20% by developing a lead scoring model using tree-based algorithms and deploying it in real-time on Kubernetes with FastAPI.

## Education

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### University of South Carolina

Masters in Computer Science — GPA: 4.0 / 4.0

Columbia, SC

01/2024 – 05/2025

Courses: Analysis of Algorithms, Compiler, Computer Architecture, Network System Security

### Indian Institute of Technology Roorkee

Bachelors in Engineering Physics — GPA: 7.03 / 10

Roorkee, India

07/2016 – 05/2020

Courses: Data Structures and Algorithms, Optimization Theory, Statistics, Linear Algebra

## Projects

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### DragAI

01/2025

- Developed an AI Knowledge Assistant that helps build RAG Applications by crawling entire websites or specific pages from URLs and storing the extracted data in vector or relational databases. The system supports multi-URL fetching, efficient indexing, and leverages a custom LLM to answer queries based on the retrieved information.

## Honors and Awards

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### 5th Big Data Health Science Student Case Competition

01/2024

- Winner of 5th National Big Data Health Science Student Case Competition for developing a model predicting sleep stages accurately.